



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CENTRE
NUMBER

INDEX
NUMBER

CLASS

H1 BIOLOGY

8875/02

Paper 2

20 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

Circle the question number in the table on the right.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 9
2	/ 9
3	/ 8
4	/ 5
5	/ 9
6 or 7*	/ 20
Total	/ 60

This Question Paper consists of **13** printed pages.

Section A

Answer **all** the questions in this section.

1. **Fig. 1.1** is a diagram of an electron micrograph of part of a plant cell.

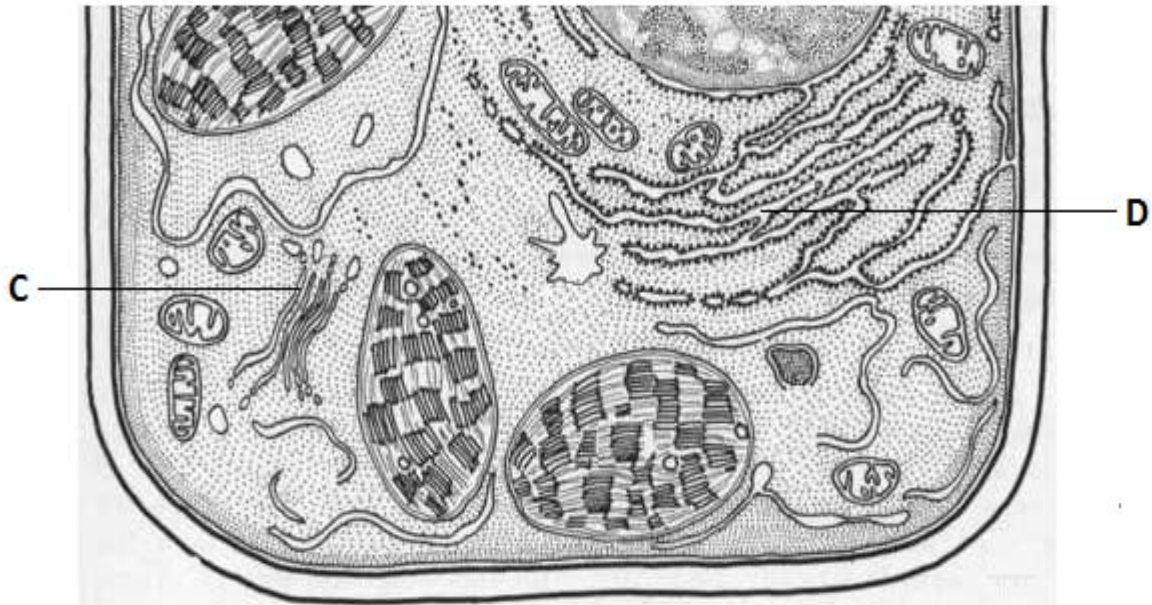


Fig. 1.1

- (a) Using lines, indicated precisely in **Fig. 1.1**, a possible location where each of the following molecules can be found:

A: chlorophyll

B: cellulose

[2]

- (b) (i) Identify the organelle labelled **C** and state a visible feature that enabled your identification.

[2]

1. Golgi apparatus
2. Made up of stacks of flattened membrane bound sacs
3. Vesicles fusing with / budding off from / associated with structure

1. (b) (ii) Explain how the functions of the two organelles, C and D, are linked. [2]

1. Vesicles containing proteins buds off from D
2. Vesicle fuses with (cis face of) C
3. releasing proteins into lumen of C
4. for further modification/ packaging/ sorting/ targeting to specific sites (any 2)

Fig. 1.2 shows a magnified image of organelle D.

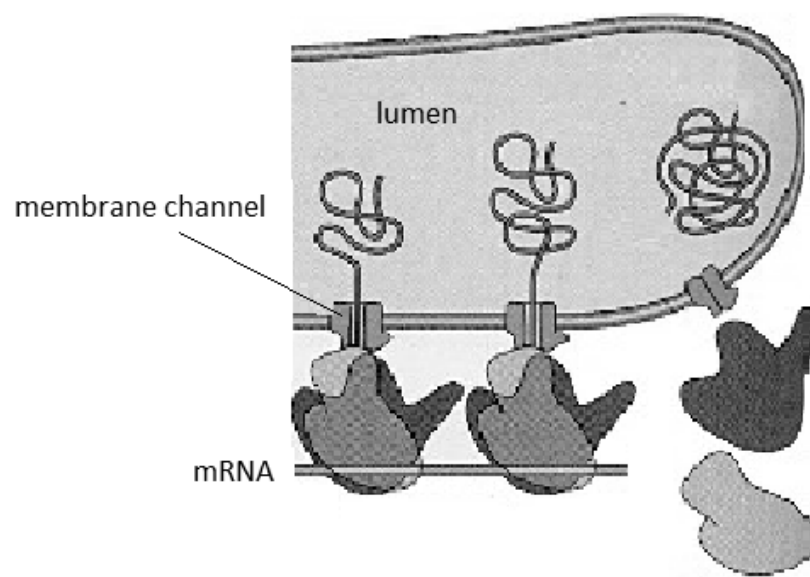


Fig. 1.2

(c) Describe the function of the protein that forms the membrane channel. [3]

1. Allow polypeptide chain to move across the membrane
2. Function as an anchor for ribosomes during protein synthesis
3. Function as a receptor to bind signal proteins to anchor ribosomes

[Total: 9]

2.

PART I

Fig. 2.1 is an electron micrograph of chromatin maintained at the 30 nm higher-order fiber.



Fig. 2.1

(a) Most of the chromatin is seen to be in the form of a fiber with a diameter of about 30 nm.

(i) Describe how DNA is packed into the 30 nm chromatin fibre. [2]

1. DNA is wound around a histone octamer
2. to form nucleosome
3. together with linker DNA made up the 10 nm chromatin fibre / nucleosome fibre
4. further coiling involved Histone H1 and
5. linker DNA to form 30-nm chromatin fibre

(ii) Explain why the genome is packaged this way. [2]

1. to regulate gene expression/ transcription
2. to make it compact/ condensed; thus DNA can be packed / contained within the cell's nucleus
3. to prevent breakage or damage; to the long DNA molecules; / to prevent the long DNA molecules from getting knotted or tangled

2.

PART II

Fig. 2.2 and **Fig. 2.3** are diagrams showing DNA replication and transcription, respectively.

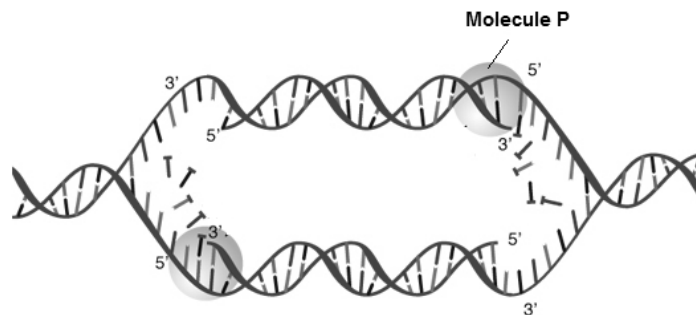


Fig. 2.2

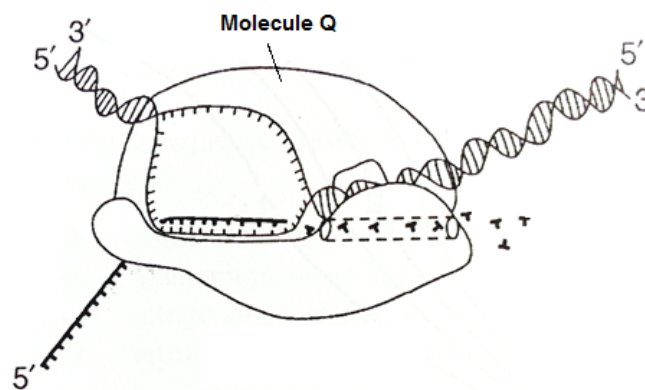


Fig. 2.3

- (b) Explain why RNA primer is required for the synthesis of new strand in DNA replication. [1]

DNA polymerase can only join nucleotides onto existing chain with free 3' OH group at one end

- (c) Describe **two** ways in which the functions of molecules **P** and **Q** are

- (i) similar to each other; [2]

Type of bond formed: phosphoester bond

Template used: DNA as template

Direction of reading the template: from 3' to 5' direction

2. (c) (ii) different from each other.

[2]

Feature	Replication	Transcription
Start site	Begins at regions of DNA known as <u>origins of replication</u>	Begins at regions of DNA known as <u>promoter</u>
Nucleotides base pair with template	Free deoxyribonucleotides/ deoxyribonucleoside triphosphate	Free ribonucleotides/ ribonucleoside triphosphate
Base pairing rule	A=T and C≡G	A=U, C≡G and T=A

[Total: 9]

3.

PART I

Fig. 3.1 is a diagram of an electron micrograph of part of a chloroplast showing thylakoid membranes.

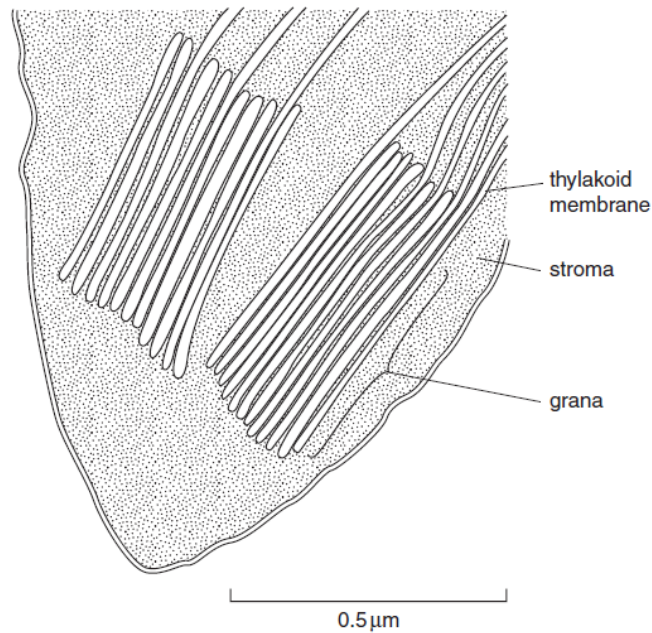


Fig. 3.1

(a) Describe the role of thylakoid membrane in photosynthesis.

[4]

1. Large surface area for embedment of
2. photosynthetic pigments for harvesting light energy
3. ATP synthase for phosphorylation of ADP to ATP
4. photosystems to generate ATP & NADPH (for Calvin cycle)
5. electron transport chain for photophosphorylation
6. Impermeable to protons
7. allow for build-up of proton gradient for chemiosmosis

3.

PART II

Arabidopsis thaliana is a small flowering plant native to Asia. A mutation in the gene coding for NADP oxidase results in plants with short root hairs. NADP oxidase is an enzyme that converts NADPH to NADP⁺. In an experiment, the length of root hairs in wild type and mutant *A. thaliana* were measured at intervals, together with the activity of NADP oxidase in the tips of the root hairs. The results are shown in **Fig. 3.2**.

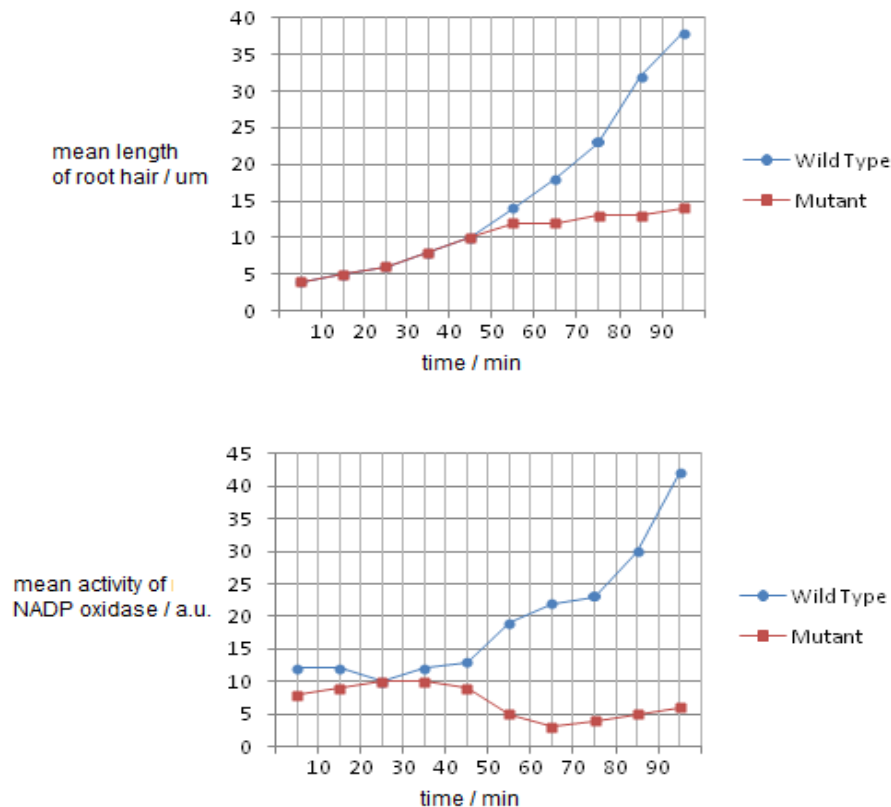


Fig. 3.2

- (b) With reference to **Fig. 3.2**, and your knowledge of photosynthesis, describe and explain the difference in the growth of root hairs in the two types of *A. thaliana*.

[4]

Describe

45 to 95 min,

1. Mean length of root hair for wild type *A. thaliana* increases from 10 μm to 38 μm
2. while that for mutant *A. thaliana* increases from 10 μm to 14 μm
3. Mean activity of NADP in wild type *A. thaliana* increases from 13au to 42au
4. while that in mutant *A. thaliana* decreases from 9au to 6au

Explain

5. NADP oxidase catalyses the conversion of NADPH to NADP⁺ for light reaction of photosynthesis

Accept: Calvin cycle of photosynthesis

6. where ATP is synthesised

Accept: synthesis of TP (if Calvin cycle mentioned in point 5)

For cell growth / growth of root hair

[Total: 8]

4. **Fig. 4.1** shows the arrangement of bones in the pentadactyl forelimbs of four vertebrates.

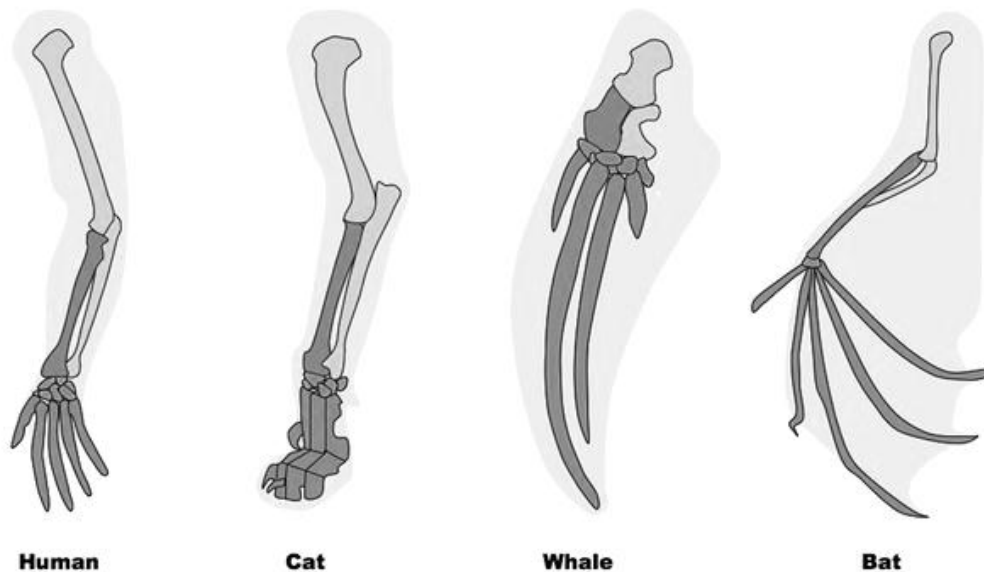


Fig. 4.1

- (a) Explain how the relationship between the structures in **Fig. 4.1** provides evidence to support the theory of evolution. [2]

1. Same arrangement of bones in the pentadactyl forelimbs of four vertebrates
2. Though they have different functions
3. Therefore they were likely to have originated from a common ancestor
4. Had been subject to different selection pressures

(b) Suggest how natural selection could have brought about the evolution of the skeleton of a bat's wing.

[3]

1. Variation in the skeletal structure
2. due to mutations/ presence of different alleles
3. Those with favourable phenotype have a selective advantage
4. They survive to reproduce
5. Pass on the allele to the offspring
6. Those at selective disadvantage will not survive
7. An increase in frequency of the allele for the favourable phenotype

[Total: 5]

5.

PART I

DNA fingerprinting is a method of identification based on an individual's DNA. In traditional DNA fingerprinting, DNA is first isolated from cells, and then amplified by Polymerase Chain Reaction (PCR). **Fig. 5.1** illustrates the process of PCR.

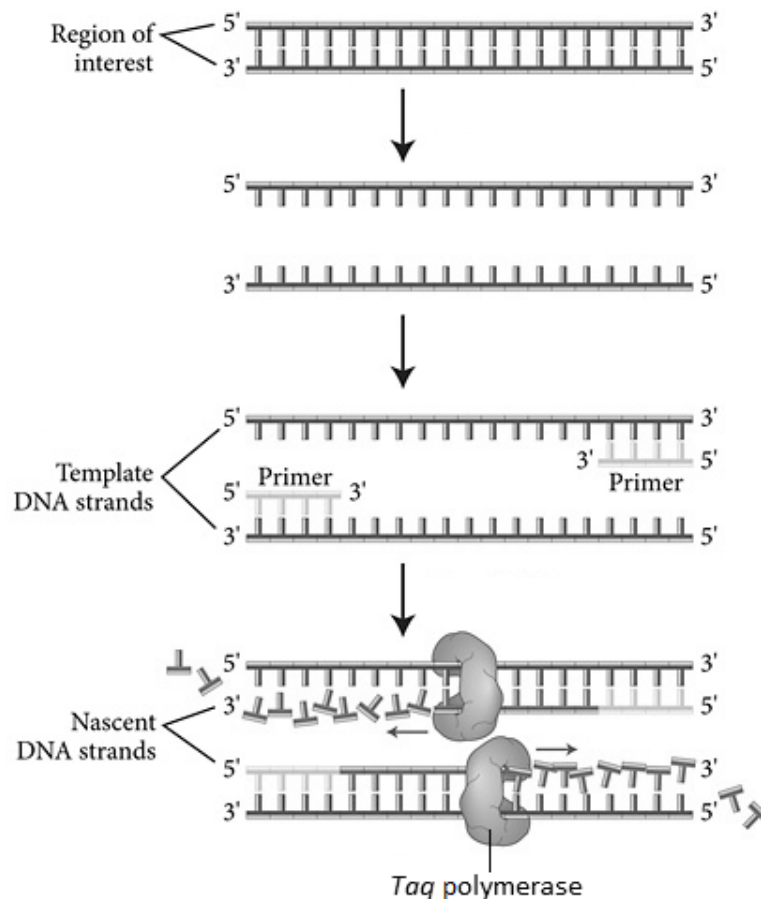


Fig. 5.1

(a) Explain how the specific nucleotide sequence of the primer is important for its role in PCR. [2]

1. Allows for selective amplification;
2. of a segment of DNA;
3. via complementary base pairing;
4. to 3' ends of the segment of interest;

5. (b) Distinguish between PCR and DNA replication in human skin cells.

[2]

Factor	PCR	DNA
Nature of primer	DNA primer	RNA primer
Enzyme for polymerisation	<i>Taq</i> polymerase	<u>Human</u> DNA polymerase
Temperature	High temperatures between 60°C to 96°C	Body temperature of 37°C
Separation of template strand	Heat	Helicase

PART II

Rice is the staple diet in many parts of the world. Rice plants produce the precursor of vitamin A, β -carotene, in their green tissues, but not in the edible endosperms of their seeds. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease.

Golden rice, which contains β -carotene, was developed in Switzerland by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium. The rice produces seeds containing β -carotene.

Fig. 5.2 shows the DNA sequence used.



Key

pro - start site for polymerase enzymes

ter - end signal for polymerase enzymes

psy - gene from daffodil (a flowering plant)

crt 1 - gene from bacterium

Hyg resist - antibiotic resistance gene from a bacterium

Fig. 5.2

The following steps were taken to produce golden rice:

- Step 1:** a length of DNA (shown in **Fig. 5.2**) was made including a rice endosperm-specific promoter, *psy* gene, *crt 1* gene and *Hyg resist* gene,
- Step 2:** copies of this length of DNA were inserted into plasmids from the bacterium *Agrobacterium tumefaciens*,
- Step 3:** *A. tumefaciens* containing the plasmids were mixed with rice embryos and cultured in a medium containing the antibiotic *Hyg*,
- Step 4:** the embryos growing successfully on *Hyg* were grown into plantlets and then plants.

5. (c) Explain how a length of DNA can be inserted into a plasmid in **Step 2**. [3]

1. use restriction, enzyme / endonuclease
2. cut plasmid / open plasmid and the length of DNA
3. sticky ends;
4. complementary bases
5. ref to anneal via hydrogen bonding / A - T / C - G
6. role of ligase, in sealing sugar-phosphate backbone / catalyses formation of phosphoester bond

(d) Explain the role of the *Hyg* resistance gene in this procedure. [1]

1. To enable selection of the transgenic cells / cells that have taken up the plasmid
2. only the cells with the new DNA can grow in the presence of the antibiotic

Field trials of genetically engineered crops, such as golden rice, hope to identify any risks to the environment or to human health of growing and eating the crops.

(e) Suggest one possible risk to the environment of growing a genetically engineered crop. [1]

- gene transfer to other rice
- contamination of other crop (e.g. ref to organic crop)
- gene transfer to wild relative
- recipient plant outcompetes others / ref to 'superweed'
- toxic to non pests

[Total: 9]

Section B

Answer EITHER 6 OR 7.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

6. (a) Describe the structure of the cell membrane and relate it to its properties. [7]

Structure	Property – Selective permeability
1. comprises of phospholipid bilayers; 2. hydrophobic tails facing inwards and hydrophilic heads facing the aqueous medium; 3. hydrophobic barrier;	P1. permeable to non-polar / hydrophobic substances;
S2a. interspersed with proteins; S2b. integral proteins are embedded in the membrane; and S2c. peripheral proteins are not embedded in the membrane;	P2. transmembrane proteins allow transport of ions/ polar/ hydrophilic substances across the membrane;
Structure	Property – Fluidity
S3. phospholipids can move laterally;	P3a. fluidity; P3b. can reseal itself after it is disrupted; P3c. can invaginate and form vesicles; P3d. transient gap for movement of small, uncharged &/ non-polar molecules;
S4. cholesterol wedged between phospholipid molecules;	P5. regulates the fluidity;
Structure	Property – Cellular identity
S5a. presence of glycolipids and glycoproteins; S5b. carbohydrates chains face the external matrix outside the cell;	P5a. membrane is asymmetrical; P5b. carbohydrate chain serves as cell surface marker / cellular recognition;

(b) Explain the significance of mitosis in growth, repair and asexual reproduction. [4]

1. Daughter cell contain the same number of chromosomes as the parent cell;
2. Due to semi-conservative replication of DNA during interphase;
3. results in genetic stability within populations of cells derived from the same parental cells;
4. daughter cells can then function harmoniously as part of the same tissue, organ and organism;

Growth

5. allows the growth and development of a zygote into a multicellular organism;
6. forming new cells that are genetically identical to the existing cell(s);

Repair

7. produces new cells required to repair worn-out or damaged tissues;
8. to return a tissue to its former condition;

Asexual reproduction

9. to produce genetically identical daughter cells that can form new offspring;
10. to reproduce rapidly in an environment that the parent plant is well adapted to;
11. colonises the habitat much more quickly than via sexual reproduction;

(c) Using named examples, explain the functions of stem cells in a living organism. [9]

1. Stem cells are unspecialised cells and has the potential to differentiate into different cell types;
2. There are 3 main types: zygote, embryonic stem cells and adult stem cells;
3. Zygote: totipotent;
4. Function: to become any cell type in the adult body and cell of extraembryonic membranes;
5. Embryonic stem cells: can be classified as early embryonic and blastocyst embryonic stem cells;
6. Early embryonic stem cells / morula: totipotent;
7. Function: to differentiate into all cell types of an individual;
8. Blastocyst embryonic stem cells: pluripotent;
9. Obtained from inner cell mass;
10. Ability to become almost any kind of cell in the body except for placenta and umbilical cord;
11. Fetal stem cells: pluripotent;
12. develop into all tissues before birth;
13. Umbilical cord stem cells: multipotent;
14. develop into a limited range of cell types in the umbilical cord;
15. Adult stem cells / somatic stem cells: multipotent;
16. undifferentiated cell found among differentiated cells in a tissue or organ;
17. The primary roles of adult stem cells in a living organism are to maintain and repair the tissue in which they are found;
18. Hematopoietic stem cells give rise to all the types of blood cells: red blood cells, B lymphocytes, T lymphocytes, natural killer cells, neutrophils, basophils, eosinophils, monocytes, macrophages, platelets (any 2);
19. Epithelial stem cells: unipotent;
20. give rise to skin cells;

[Total: 20]

7. (a) Distinguish between competitive and non-competitive inhibition of enzymes.

[7]

	Competitive Inhibition	Non-competitive Inhibition
1	Structural similarity to substrate	No structural similarity to substrate
2	Binds to active site	Binds to site other than active site
3	Usually binds temporarily / inhibition usually reversible	Binds temporarily or permanently / inhibition can be reversible or irreversible
4	Works by competing with substrate for active site Decreasing proportion of active sites available for binding to substrate	Works by binding to a region other than the active site and inducing change in conformation/shape of the enzyme including that of the active site Thus rendering active site less receptive to substrate / less effective in catalysing conversion of substrate to product
5	Prevents / lowers frequency of formation of ES complexes	May or may not prevent formation of ES complex
6	Does not change conformation of enzyme upon binding	Changes conformation of enzyme upon binding
7	Inhibition can be overcome by high substrate concentration	Inhibition cannot be overcome by high substrate concentration
8	Maximum reaction rate upon increasing substrate concentration is the same with or without inhibitor	Maximum reaction rate upon increasing substrate concentration is lower with inhibitor
9.	AVP	

(b) Describe how the Calvin cycle differs from the Krebs cycle. [8]

	Feature	Calvin cycle	Krebs cycle
1	Location	Stroma of chloroplast	Mitochondrial matrix
2	Nature of process	Anabolic	Catabolic
3	Carbon dioxide	Fixed with RuBP	Released by oxidative decarboxylation of respiratory substrate
4	Enzymes involved	Rubisco, which catalyses carbon fixation	Dehydrogenases, which catalyses oxidation
5	Coenzymes involved	Reduced NADP	Reduced NAD and FAD
6	Role of coenzyme	NADPH / H^+ reduces / donates electrons to 1,3-bisphosphoglycerate	Oxidised NAD and FAD accepts electrons from respiratory substrates
7	Fate of substrate	Reduced	Oxidised
8	Fate of ATP	Used to activate 3-phosphoglycerate	Produced by substrate level phosphorylation
9	AVP		

(c) The genetic manipulation of plants and animals, and their use in agriculture, is one the most controversial scientific development of recent times. What are some moral issues associated with the rise of such technology? [5]

1. Violation of natural organism's intrinsic values with the mixing of genes among species / tampering of genes
2. Animal welfare could be adversely affected due to little concern about whether animals could withstand the additional burden of increased milk / meat production
3. Controversies on patent of GMO as patent of transgenic plants / animals reduces them to the level of objects
4. Religious / dietary concerns with consuming animal genes in plants / relevant examples
5. Labelling of GM food is not mandatory in some countries thus lack of awareness for those who feel strongly against consumption of foreign DNA incorporated into their food
6. AVP

[Total: 20]